

Month 3: Dense Representation Learning and Vector Ingestion

This month focused on dense embeddings, vector database index setup, and the implementation of the first-stage rank-merging node.

Week 1: Embedding Model Selection and Local Setup

- **Progress Summary:** The local BAAI/bge-m3 embedding model was selected for dense vector representation. The model was configured using the HuggingFace transformers library, and local CPU and RAM utilization were profiled to optimize encoding performance.

Week 2: Vector Database Connector Development

- **Progress Summary:** The vector database manager class (VectorStoreManager) was implemented. Connection adapters were built to interface with the Pinecone Cloud API. The code was configured to parse API keys and environment variables from a secure .env file.

Week 3: Batch Vectorization and Ingestion

- **Progress Summary:** The batch vectorization pipeline was executed. Wikipedia text chunks were converted into 1024-dimensional dense vectors using BGE-M3 and uploaded to Pinecone in batches of 100 to optimize network throughput and avoid API throttling.

Week 4: Rank Merging and Deduplication

- **Progress Summary:** A rank-merging node was designed to consolidate results from the lexical and dense paths. An integration test was executed: queries were run through the local BM25 search and the Pinecone dense search, and the outputs were merged and deduplicated.